Constructing the graph, then confronting it with data

STAGE 1–4 · no data consulted

1. Nodes

25 constructs, grouped by causal role (hierarchical abstraction)

2. Edges

53 papers screened on temporal precedence, mechanism, empirical support

3. Expert review

interactive DAG + issue tracker, then workshop sign-off → 89 edges

4. Identification

backdoor criterion (dagitty)
$\mathcal{A}$ = {Climate, Econ. Capital, Housing, Tenure, Urbanicity}

the graph is fixed here

STAGE 5 · data enters (n = 31,802 LSOAs)

5a. Test the implied independencies

80 implications of the 16-node measured subgraph, judged by |partial r|, each labelled by whether the pair is d-connected in the full 25-node DAG

5b. Recover structure from data alone

PC-stable (assumes no latent confounders) and FCI (latent-robust), $\alpha \in \{.01, .005, .001\}$, with and without the expert ordering as background knowledge

Verdict against pre-stated conditions

(i) no HEE–MH adjacency appeared; (ii) failed implications were dominated by latent-cut pairs, not by pairs the full DAG separates. Neither triggered.

$\mathcal{A}$ carried forward to the companion estimation paper

What stage 5 cannot do

Discovery returns an equivalence class, so a recovered orientation is not evidence of causal direction. It cannot estimate the HEE → MH effect: all but one of the mediators carrying it are unmeasured. Agreement therefore corroborates the graph; only disagreement of the two pre-stated kinds could refute it.